Reg. No. | | | | | | | | | | [13CE61/12CE61]

SIXTH SEMESTER B.Tech./B.E EXAMINATION, JUNE – 2016

DESIGN OF STEEL STRUCTURES

Time: 3 Hours **Maximum Marks: 70**

Instructions:

1. Answer any **FIVE** full Questions.
2. Missing data may be suitably assumed.
3. IS 800 : 2007 and Steel Tables are permitted.

1. a. Explain the types of bolted connections with neat diagrams. **04**

b. Determine the bolt valve of 20mm diameter bolt, connecting 10mm plates in single shear and also in double shear. Bolts used are 4.6 grade and plate is of 410 grade. Take area of bolt as 245mm^2. **10**

2. a. What are the advantages of welded connections **04**

b. A bracket transmits a load of 20kN at an eccentricity of 30mm through 6 bolts of M20 bolts and of 4.6 gade, arranged in two vertical rows 60mm apart as shown in figure1. The pitch of the bolt is 60mm and the total load lies in the plane of the joint . Determine the strength of the bolt and the maximum stress in the bolt. The grade of steel plate is 410. **10**

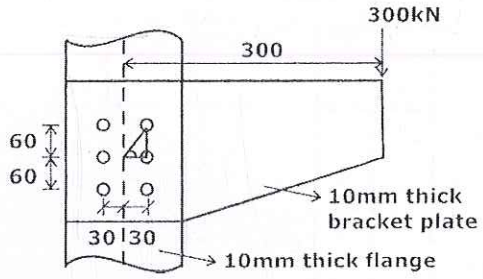

Fig. 1 (all dimensions are in mm)

3. a. What are the advantages of bolted connections. **04**

b. Determine the bracket load for the connection shown in fig. 2. Use M20 bolts and 4.6 grade. **10**

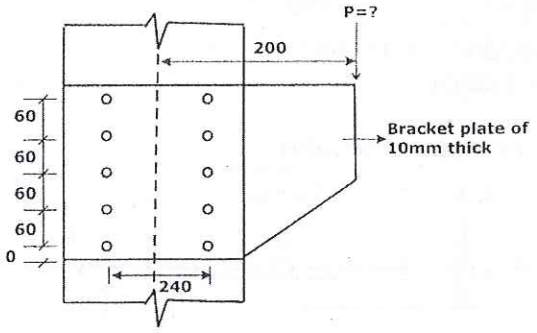

Fig. 2 (all dimensions are in mm)

4. A tie member of a roof truss consists of 2 ISA100mm×75mm×8mm. The angles are connected to either side of a 10mm gusset plates and the member is subjected to a working pull of 300kN. Design the welded connection. Assume connections are made in the workshop. **14**

5. Design a column section consisting of 4 angle section arranged to a box shape 400mm×400mm to carry an axial load of 2500kN. The height of the column is 5m and both the ends are fixed. **14**

6. A simply supported steel joist of 4.0m effective span is laterally supported throughout. It carries a total uniformly distributed load of 40kN inclusive of self weight. Design an appropriate section using steel of grade Fe410. **14**

7. Design a bridge truss diagonal subjected to a factored tensile load of 300kN, the length of the diagonal is 3.0m. The tension member connected to a gusset plate 16mm thick with one line of 20mm diameter bolts of grade 8.8. **14**

8. Design the column bracket to support a gantry girder, following the arrangement shown in fig 3 and fig 4.

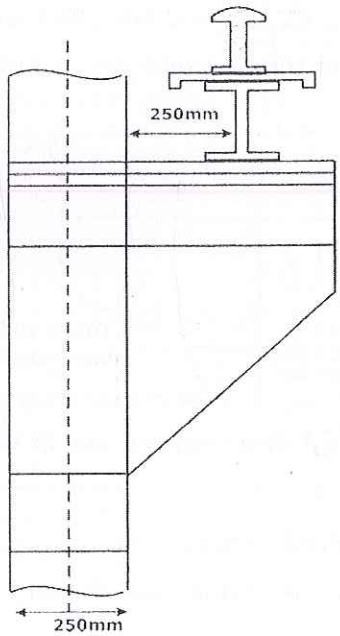

Fig. 3

Where, span of gantry girder = 6 meters
Capacity of the crane = 100kN
Wheel base = 3m
Maximum load on each carriage = 97.5kN

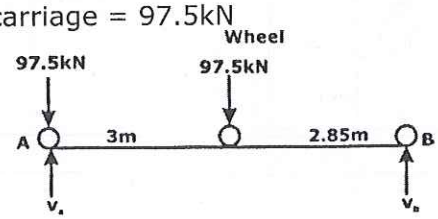

Fig. 4